

# Hurricane Forecasting Report

## A. Feature & Labels

### 1. Input Feature

The original ML 2026A2 Hurricane dataset contained six variables

- rain precipitation rate
- wind speed
- surface air pressure
- downward short-wave radiation flux
- downward long-wave radiation flux
- wind direction

The wind direction is represented in degrees, and directional data are circular, meaning  $0^\circ$  and  $360^\circ$  represent the same direction. So, using degree values directly could produce discontinuities and negatively affect the model learning. Therefore, wind direction was converted into sine and cosine components to preserve directional continuity and improve model learning.

| <i>File given</i> | <b>Features</b> | <b>Description</b>                 | <b>Unit</b>                          |
|-------------------|-----------------|------------------------------------|--------------------------------------|
| Rainf_tavg        | rain            | Rain precipitation rate            | Kilogram per square metre per second |
| Wind_f_inst       | wind_spd        | Wind Speed                         | Metre per second                     |
| Psurf_f_inst      | pressure        | Surface air pressure               | Pascal                               |
| SWdown_f_tavg     | sw_down         | Downward short-wave radiation flux | Watt per square metre                |
| LWdown_f_tavg     | lw_down         | Downward long-wave radiation flux  | Watt per square metre                |
| Wind_dir_p1000    | wind_dir_sin    | Sine-transformed wind direction    | Degree                               |
|                   | wind_dir_cos    | Cosine-transformed wind direction  |                                      |

Table 1. Input feature details

For model training, seven weather input features were extracted from the hurricane dataset. This was because these features are strongly associated with hurricane development, behavior and storm intensity.

## 2. Input Dimensionality

The model was trained to learn temporal weather patterns from historical observations in which the weather data from a fixed number of previous days was used to predict the future atmospheric conditions.

A 14-day historical observation window was selected as the input sequence length. For each training sample, the model receives observations from the previous 14 consecutive days. The purpose of using the 14-day window was to ensure that the model could identify any short-term trends, atmospheric changes and changing weather conditions associated with hurricanes over time. Hurricanes typically evolve slowly instead of occurring spontaneously, so using multiple previous days provides the model with more information compared to using a single day observation.

Since the input sequence window contained 14 days, and each day has 7 input features, the total input dimensionality becomes:

$$14 \text{ days (INPUT\_DAYS)} * 7 \text{ input features} = 98$$

Therefore, each training sample consisted of 98 numerical input values.

The forecasting was set to 7 days to predict hurricane-related atmospheric conditions one week into the future using the previous 14 days of observations. A 7-day forecasting horizon was selected because it offered an achievable medium-term weather prediction problem where the model would have enough time to extract meaningful temporal relations from past to future atmospheric conditions.

Since multilayer perceptron models only accept vector inputs, the produced temporal sequences were flattened to a one-dimensional vector form before they could be fed into the multilayer perceptron model for processing.

## 3. Feature Extraction

The dataset contained weather observations recorded every three hours for several locations. To generate daily observations suitable for forecasting, the measurements were averaged across each day.

As mentioned previously, wind direction was transformed into sine and cosine components using a trigonometric transformation that produces directional continuity and enables the multilayer perceptron model to learn wind circulation patterns more effectively.

To stabilise the model training further, rainfall attributes were transformed to a log function to reduce the skewness caused by extreme precipitation events. Numerical weather variables were standardised to place features on similar scales, and extreme outlier values were clipped to reduce instability during training.

## **4. Output Labels**

The model was designed as a multivariate forecasting system that predicts future hurricane-related atmospheric conditions seven days ahead. The target labels were extracted using a forecasting horizon of 7 days: rain, wind speed, wind direction sine, wind direction cosine, and pressure. Since the model predicts five meteorological variables, the output dimensionality was 5. The forecasting task involves four physical meteorological parameters. Sine and cosine functions are used to avoid the false discontinuity between 359 deg and 0 deg. Thus, the model performs multivariate time-series forecasting for multiple hurricane-related atmospheric conditions.

## **B. Preprocessing**

Several preprocessing steps were applied to prepare the hurricane dataset for training the multilayer perceptron forecasting model.

### **1. Daily Aggregation**

The original dataset contained meteorological observations recorded every three hours. For forecasting purposes, it was necessary to have daily observations by averaging eight measurements taken in a day, a single observation per day was obtained for each meteorological parameter.

### **2. Data Merging**

The variables from each dataset were then combined into a single dataframe using the same attributes such as year, month, day, latitude and longitude. This produced one unified dataset containing all weather variables for each location and date.

### **3. Missing Value Handling**

The missing values from wind direction were replaced with zero before transformation to prevent invalid numerical values during model training.

### **4. Rainfall Log Scaling**

Rainfall values were transformed using the logarithmic function  $\log_{1p}(\text{rain} \times 86400)$ . The multiplication by 86400 roughly translated the rain rate to a daily scale before applying log scaling. This conversion minimised the skewness due to the outliers in rain rate, normalised high rain rates, and stabilised the computations in neural network training.

### **5. Feature Normalisation**

The variables include rain, wind speed, pressure, short-wave radiation flux and long-wave radiation flux, which were standardised using z-score normalisation. This ensured that variables shared similar numerical scales, improving optimisation stability during training.

### **6. Outlier Clipping**

After normalisation, numerical values were clipped to the range  $[-5, 5]$  to reduce the influence of extreme outlier values and improve numerical stability during training.

## 7. Dataset Splitting

The generated sequences were divided chronologically into training, validation, and testing datasets to prevent future information leakage. The dataset split was

- Training data: up to 31 December 2023
- Validation data: 1 October 2024 to 10 December 2024
- Testing data: from 11 December 2024 onward

A chronological split was used instead of random sampling because hurricane forecasting is a time-series prediction problem where future observations must remain unseen during training.

## C. Model

### 1. Model Architecture

The MLP network model for hurricane prediction was constructed based on the feed-forward Multi-Layer Perceptron (MLP) approach using the PyTorch framework. The model was a regression-based network that forecasts weather conditions in the upcoming seven days based on previous weather data.

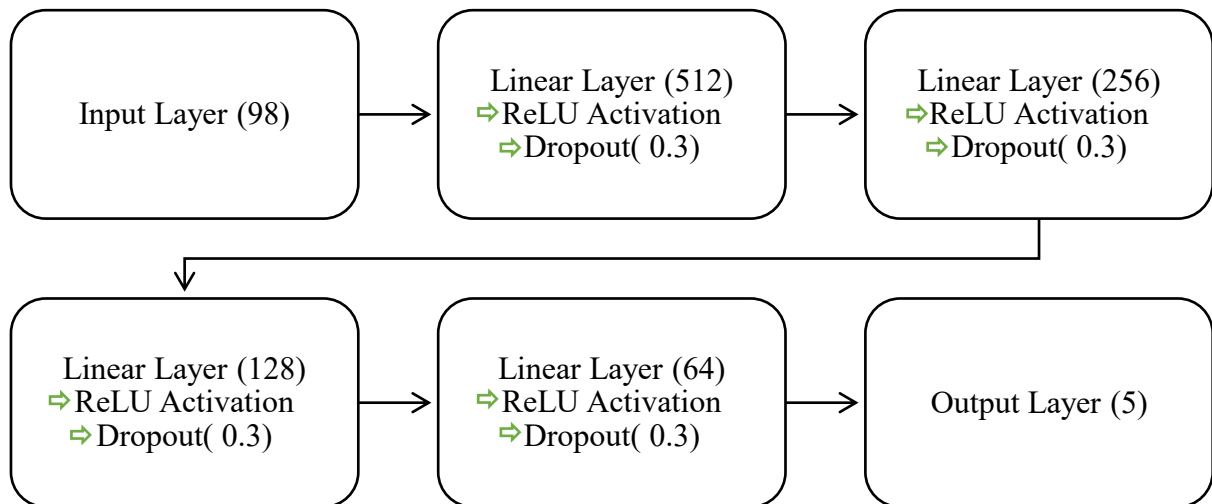


Figure 1. Architecture of the implemented PyTorch multilayer perceptron model.

Figure 1 illustrates the architecture of the implemented multilayer perceptron forecasting model. The network consisted of four hidden fully connected layers with ReLU activation and dropout regularisation, followed by a five-dimensional regression output layer. The output layer produced five regression outputs that are rainfall, wind speed, wind direction sine, wind direction cosine and surface pressure.

## 2. Hyperparameters

The following hyperparameters were used during training

| Hyperparameters         | Values             |
|-------------------------|--------------------|
| Input sequence length   | 14 days            |
| Forecast horizon        | 7 days             |
| Input dimensionality    | 98                 |
| Output dimensionality   | 5                  |
| Hidden layer sizes      | 512, 256, 128, 64  |
| Dropout rate            | 0.3                |
| Learning rate           | $1 \times 10^{-4}$ |
| Weight decay            | $1 \times 10^{-4}$ |
| Batch size              | 128                |
| Maximum epochs          | 30                 |
| Early stopping patience | 10                 |

Table 2. Hyperparameters details

## 3. Model Optimisation

The model was optimised using the Mean Squared Error (MSE) loss. Mean Squared Error was selected because the task involved multivariate regression forecasting, where the objective was to minimise the difference between predicted and actual meteorological values. A ReduceLROnPlateau learning-rate scheduler was also applied during training. The scheduler automatically reduced the learning rate when validation loss stopped improving, allowing more stable optimisation during later training stages.

### Rationale for the Multilayer Perceptron

A multilayer perceptron was selected because the forecasting task required learning nonlinear relationships between multiple meteorological variables and future atmospheric conditions. The temporal sequences allowed the model to learn interactions between historical weather observations and future hurricane-related behaviour. The progressively decreasing hidden-layer dimensions enabled the model to gradually compress learned representations while retaining important forecasting information. The multilayer perceptron also offered a simple and computationally efficient method for prediction modelling in meteorological data.

## 4. Overfitting Prevention

- Dropout Layering – Preventing the network from relying too heavily on specific features and reducing co-adaptation between hidden units.
- Weight Decay Regularisation – To optimise excessively large model weights and encourage simpler model representations.
- Early Stopping – To prevent the model from continuing to learn noise from the training dataset after generalisation performance had stabilised.

These overfitting-prevention techniques improved the model's ability to generalise to unseen hurricane-related weather observations.

## D. Generalizability

The final prediction dataset generated during testing contained 32,787 supervised forecasting instances. To properly assess model generalisation, the dataset was divided chronologically into training, validation, and testing subsets.

A chronological time-based split was used rather than random sampling. This approach was more appropriate for hurricane forecasting because the model was evaluated on future atmospheric conditions that were not observed during training, better reflecting realistic forecasting scenarios. The generated temporal sequences were divided according to their target dates. Training sequences contained target dates up to 31 December 2023. Validation target dates ranged from 1 October 2024 to 10 December 2024, while testing target dates began from 11 December 2024 onward. A temporal split was employed because it would minimise information leakage in future, which may occur when employing random splitting while predicting time series models. The fact that 14 days of data were used in observing the trend also meant that the split generated a time buffer around each dataset.

After training, the best-performing model checkpoint was loaded and evaluated on the unseen test dataset. Predictions generated for each mini-batch were collected and combined into a complete prediction and target arrays.

The performance values were computed by directly comparing the predicted meteorological variables with the true target values from the unseen test dataset. The Mean Squared Error measures the average squared difference between predicted and actual atmospheric values, while the Mean Absolute Error measures the average absolute prediction error. Lower values for both metrics indicate that the model predictions were closer to the actual hurricane-related weather observations.

Mean Squared Error is given by

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

And Mean Absolute Error is

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

The final forecasting performance of the multilayer perceptron model on the unseen test dataset is

- Test Mean Squared Error (MSE): 0.4332
- Test Mean Absolute Error (MAE): 0.4803

The model demonstrated the ability to learn meaningful relationships between historical meteorological observations and future atmospheric conditions. The use of chronological splitting, dropout regularisation, weight decay, validation monitoring, and early stopping

improved the model's ability to generalise to unseen hurricane-related weather observations. The reported test metrics, therefore, reflect the forecasting performance of the model on previously unseen atmospheric conditions rather than memorised training data.